

Design Thinking, Innovation and Entrepreneurship (25HU102)

Question Bank

Unit-1 Short answer questions

1. Define Design Thinking.
A human-centered problem-solving approach that combines creativity, empathy, and logic to create solutions that meet real human needs.
2. State any two reasons why Design Thinking matters.
 - (i) Solves complex problems in innovative ways.
 - (ii) Improves products, services, and experiences.
3. List any two key characteristics of Design Thinking.
Human-centered, Creative (also Iterative, Collaborative).
4. How does Design Thinking put people first?
By ensuring solutions are based on real human needs, not assumptions.
5. Give two ways in which Design Thinking encourages innovation.
Promotes out-of-the-box thinking and helps discover fresh solutions to complex problems.
6. Why is Design Thinking useful for solving complex problems?
It helps break “wicked problems” into manageable, testable steps.
7. How does Design Thinking reduce risk of failure?
Through early prototyping, testing, and feedback-driven improvements.
8. Mention two industries where Design Thinking is applied.
Healthcare, Government (also IT, Education, Business, Social Projects).
9. List two essential skills built by Design Thinking.
Creativity, Empathy (also Collaboration, Adaptability).
10. Who introduced the idea of ‘wicked problems’? In which year?
Horst Rittel & Melvin Webber, 1970.
11. Name the book written by Herbert A. Simon and its year.
The Sciences of the Artificial, 1969.
12. Who expanded the concept of Design Thinking in the 1980s?
Rolf Faste (Stanford).
13. Who founded IDEO and in which year?
David Kelley, IDEO formed in 1991.
14. Which institute launched Design Thinking as a 5-stage process in 2005?
Stanford d.school (Hasso Plattner Institute of Design).
15. Name the 5 stages of Design Thinking proposed by Stanford d.school.
Empathize, Define, Ideate, Prototype, Test.
16. Who published *Change by Design* and when?
Tim Brown (CEO of IDEO), in 2009 onwards.
17. Which two pioneers laid early foundations of Design Thinking in the 1950s–60s?
John E. Arnold (Stanford) and Herbert A. Simon.
18. Give one contribution of Robert McKim to Design Thinking.
Introduced visual thinking and integrated problem-solving in design education (1970s).
19. Mention any two sectors where Design Thinking is widely adopted in the 2010s–present.
IT, Government Policy (also Healthcare, Education, Social Innovation)
20. State one major difference between Conventional Problem-Solving and Design Thinking.

21. What is the focus of Design Thinking?
The focus of Design Thinking is human-centered problem-solving, i.e., understanding people's needs first.
22. List the five stages of the Design Thinking process.
Empathize → Define → Ideate → Prototype → Test.
23. On what basis does conventional problem-solving rely compared to design thinking?
Conventional problem-solving relies on logic, past data, and expert opinion, whereas design thinking relies on empathy, experimentation, and user feedback.
24. How is failure treated in Design Thinking?
Failure is treated as a learning opportunity through quick tests and iterations.
25. What type of innovation is common in Design Thinking?
Design Thinking often leads to innovative and disruptive solutions.
26. Mention one key advantage of multidisciplinary teamwork in Design Thinking.
It fuels innovation by bringing diverse perspectives for richer, well-rounded solutions.
27. Which company reduced product development cycle time by 33% using Design Thinking?
IBM.
28. What is the focus of Design Thinking?
The focus of Design Thinking is understanding and meeting people's needs (human-centered approach).
29. How did Nike apply Design Thinking to improve its products?
Nike used empathy research to design Flyknit shoes, which are lightweight, eco-friendly, and athlete-focused.
30. What problem did Airbnb solve using Design Thinking?
Airbnb redesigned its booking process and trust-building features after interviewing hosts and guests, helping it grow from near-bankruptcy to success.
31. Name the healthcare organization that improved patient waiting areas through Design Thinking.
Mayo Clinic.
32. Which FMCG company created Swiffer through Design Thinking?
Procter & Gamble (P&G).
33. How did Ford enhance customer experience using Design Thinking?
Ford used rapid prototyping and user testing to develop the SYNC in-car connectivity system, improving customer experience and loyalty.
34. Name the banking initiative developed by Bank of America using Design Thinking.
"Keep the Change" program.
35. Write one short difference between the outcomes of Conventional Problem-Solving and Design Thinking.
Conventional problem-solving → arrives at one best solution chosen early.
Design Thinking → tests and refines multiple ideas for user-centered solutions.
36. Why is it important to include a diverse mix of participants in design thinking workshops?
Because diversity brings different perspectives, reduces bias, and prevents "groupthink," leading to richer ideas and solutions.
37. Name three physical tools and two digital tools that can be used in workshops.
Physical tools: Sticky notes, markers, whiteboards.
Digital tools: Miro, FigJam.

38. What is the function of a facilitator in a design thinking meeting?
The facilitator keeps the session on track, encourages participation, and ensures objectives are met.
39. Compare the role of a facilitator and a scribe in workshops. Why are both important?
40. Facilitator: Guides the process, keeps participants engaged, and maintains focus.
41. Scribe: Records ideas, decisions, and outcomes for documentation.
Both are essential because one manages process flow while the other ensures knowledge retention.
42. What are the three stages of the IDEO Design Thinking model?
43. Who can be included in a multidisciplinary Design Thinking team? (Any two)
44. What is the role of a Facilitator in a Design Thinking team?
45. Which team role is responsible for managing user research and empathy work?
46. Who is responsible for turning ideas into tangible forms in a Design Thinking team?
47. State any two team norms essential in Design Thinking collaboration.
48. How does collaboration speed up problem-solving in Design Thinking?
49. Mention any two key characteristics of Design Thinking.
50. Name the book written by Herbert Simon in 1969 that laid the foundation for Design Thinking.
51. Which institute launched the academic program of Design Thinking as a 5-stage process in 2005?
52. Who published *Change by Design* in 2009?
53. Expand IDEO and name its founder.

Long answer questions

1. Apply the 5-step process of Design Thinking to the following case study.
 1. Mumbai Dabbawallas
 2. Jeevan Bindi
 3. Tata Nano Car
2. Explain the importance of Design Thinking in solving complex problems with suitable points.
3. Describe the key characteristics of Design Thinking.
4. Write a short note on the history of Design Thinking (any four key milestones).
5. Why is Design Thinking considered a human-centered approach? Justify with examples.
6. Discuss how Design Thinking reduces the risk of failure in innovation.
7. Explain the contributions of Herbert Simon, Robert McKim, and Rolf Faste to Design Thinking.
8. Define Design Thinking and explain its importance in solving complex, real-world problems.
9. Compare Conventional Problem-Solving and Design Thinking with respect to approach, focus, and outcome.
10. Why is collaboration more effective in Design Thinking compared to conventional problem-solving?
11. How did Nike use empathy in Design Thinking to develop Flyknit shoes? What was the result?
12. Describe the role of Design Thinking in Airbnb's business transformation.
13. How did the Mayo Clinic improve patient experience using Design Thinking?

14. Explain the process through which Procter & Gamble (P&G) created the Swiffer product line using Design Thinking.
15. Trace the history of Design Thinking. Highlight at least five major milestones and contributions of key thinkers.
16. Explain why Design Thinking is considered human-centered. Support your answer with examples from industries like healthcare, education, or business.
17. "Design Thinking reduces the risk of failure." Justify this statement with examples and reasons.
18. How has Design Thinking evolved from the 1950s to the present day? Explain with reference to important contributors such as Herbert Simon, Rolf Faste, and IDEO.
19. What are 'wicked problems'? How does Design Thinking provide effective solutions for such problems?
20. Evaluate the applications of Design Thinking across different fields such as business, education, healthcare, and government.